

Reporting Degenerate Emissions in a 135M Diffusion Language Model: A Reproducible Negative-Control Pilot Case Study

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Abstract

This reproducible pilot case study examines a negative control on one 135M-parameter masked diffusion language model bed. It is confined to artifact reproduction and engineering diagnosis in this bed, and claims no new evaluation metric, training method, or discovery. Under a matched short budget, the last-logged loss favors autoregressive over random initialization by 2.26 nats, in the same direction across four paired data-order seeds; the random initialization itself is one fixed draw, so these repeats do not measure weight-draw variance. In a single-prompt generation pilot both arms are visibly degenerate, yet teacher perplexity ranks them opposite to the paired loss benchmark while effective length, alphabetic yield, and emitted-token concentration separate their emissions. Without an independent quality anchor, neither observation establishes metric capability or a quality ranking. MAUVE is *not recorded* under this pilot configuration, so we form no ratio and draw no MAUVE capability or quality inference. The engineering output is instead a companion record for this pilot: place effective emission length and emitted-token degeneration beside per-token scores. It is not a new scalar metric, and nothing here extrapolates beyond the stated bed and pilot.

1 Introduction

Masked diffusion language models generate by repeatedly denoising a canvas of masked tokens rather than extending a causal prefix one token at a time (Sahoo et al., 2024; Nie et al., 2025). This design exposes intermediate signals—entropy, maximum probability, mask state, and changes in the current argmax—that can guide adaptive decoding (Li et al., 2025). Generated-text PPL, MAUVE, and trajectory summaries describe different projections of an emission; none is an independent human-quality anchor in our experiment. Wang et al. diagnose length, repetition, and punctuation as factors that can distort PLM-based perplexity (Wang et al., 2022). We therefore ask only whether two known-degenerate outputs in one small dLLM pilot differ on analogous emission fields that our initial report did not place beside each score.

This paper turns a small training bed into a controlled negative-control pilot case study. The bed starts from SmolLM2-135M (Ben Allal et al., 2025) and converts the same architecture to a masked denoiser. One arm reuses autoregressive weights; the other uses a fixed random-initialization checkpoint. Data, optimizer, schedule, token count, hardware, sampler, and evaluation are held constant. This design does not estimate performance over the space of random initializations. It asks a narrower question: at a fixed 30.0M-token budget, how do the two degenerate arms differ on directly audited emission fields when the initialization is replaced by a constructive negative control?

Three categories are easy to conflate. *Reproduction* means that the reported arithmetic and artifacts can be rerun; it is not a multisite replication. *Engineering diagnosis* means applying a transparent audit to this bed. A *genuinely new method* would require an independently validated metric, training rule, or decoder. This paper is a reproducible negative-control diagnosis in the first two categories, not the third. Its three concrete deliverables are:

1. A matched-budget calibration shows a same-signed loss separation over four paired data-order seeds: the mean last-logged loss gap is 2.262, with 95% CI [2.220, 2.305]. The result is explicitly conditional on the two fixed initialization checkpoints and short 458-step horizon.

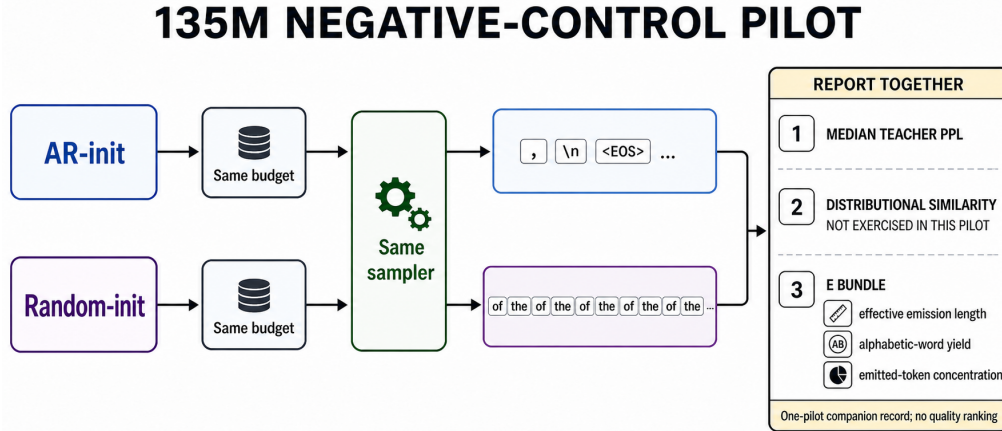


Figure 1: **Companion record for this single 135M pilot — the slots of Eq. (1).** Both arms are degenerate: AR-init stops after punctuation/newline remnants, whereas random-init continues as a repeated word stream. The panel matches Eq. (1) exactly: (i) median teacher PPL; (ii) distributional similarity, *not exercised* here because the attempted MAUVE measurement is not recorded under this configuration (Section 4.2); and (iii) $\mathcal{E}$, comprising effective emission length, alphabetic-word yield, and emitted-token concentration. No ratio or quality inference is made on slot (ii). Argmax concentration is intentionally excluded because the final argmax and emitted distribution decouple in our data (Section 4.2). This descriptive audit accompanies existing measurements; it is neither a new scalar metric nor a quantitative result, and all measured values remain in the tables.

2. In this single pilot, a report-level audit shows that the two degenerate arms differ in effective emission length and emitted-token degeneration. These fields are not part of the reported PPL value, while the attempted MAUVE value is not recorded; we place them beside each PPL cell only for this setting, not as a general requirement or new metric.
3. A frozen signal schema and reproducible scripts record the denoising trajectory. Unigram, drift, sweep-tail, and cross-bed probes are engineering diagnostics rather than a proposed decoder or training method.

The central claim is deliberately modest: both arms are degenerate and differ on emission coordinates outside this pilot’s PPL entry; the attempted MAUVE entry is not recorded, and no independent assessment supplies a quality ranking. Figure 1 summarizes the resulting companion record for this single bed, not a new evaluation method or a statement about other prompts or dLLMs.

2 Related Work and Evaluation Framing

Masked diffusion language modeling. Masked diffusion objectives can be written as mixtures of masked-language modeling losses and paired with iterative unmasking samplers (Sahoo et al., 2024). LLaDA demonstrated that the paradigm scales to an 8B parameter model trained from scratch (Nie et al., 2025); more recent systems span reusable training frameworks (Zhou et al., 2026), continuous embedded flows (Hu et al., 2026), and adaptive generation length (Li et al., 2025). AR-to-diffusion adaptation has already been studied systematically in the DiffuLLaMA family (Gong et al., 2025); PreDiff-LM studies pretrained masked diffusion with hybrid attention (Yao et al., 2026), and UNIFUSION studies AR adaptation under a unified reverse-rate objective (Jiang et al., 2026). Accordingly, our matched initialization arms are a small reproducible case study, not a new adaptation method or training discovery.

Metrics score different projections of emitted text. Generated-text PPL scores teacher likelihood; Wang et al. show that length, repetition, and punctuation can distort that score (Wang et al., 2022). MAUVE instead compares generated and human distributions (Pillutla et al., 2021); surface form also matters. Final-argmax share describes concentration of the generator’s decisions. Within the 11-source local verification ledger used for this paper, we did not establish that these sources

Table 1: **Scope comparison with the closest locally verified work.** NR means that a field was not measured in this implementation or was not established by the frozen local verification ledger; no value is inferred from memory. Relative to Wang et al. (2022), this pilot audits the same confounding factors on a matched-initialization dLLM bed and adds the within-arm check in Table 6; no novelty claim rests on an NR cell.

Work	Model scale	Training budget	Prompts	Reported evaluation	Human anchor	Emission reporting
Wang et al. (2022)	PLM-PPL study	Not a generator-training comparison	NR	PPL; analyzes length, repetition, punctuation	NR	NR
Gong et al. (2025)	DiffuLLaMA 7B family	NR	NR	AR-to-diffusion adaptation study	NR	NR
Yao et al. (2026)	PreDiff-LM; NR	NR	NR	Pretrained masked diffusion with hybrid attention	NR	NR
Jiang et al. (2026)	UNIFUSION NR	NR	NR	Unified reverse-rate AR adaptation	NR	NR
This pilot	135M	30.0M tokens/arm; 458 steps	1	Two-teacher PPL, MAUVE, argmax and trajectory probes	None	Length, alphabetic yield, top-word share

report whether the emitted object is a short EOS stub or a long lexical loop. Our fields expose those factors on one dLLM bed. A within-arm, within-prompt before/after check (Section 4.2) establishes complementarity with teacher PPL only for that pair; discriminative evidence across initializations and prompts remains unresolved. Here PPL ranks the arms opposite to the matched four-seed loss benchmark, but the pre-registered multi-init control reproduces only the loss-side sign while its degeneration mode varies by draw (Section 6). The conflict therefore motivates the audit without establishing a quality anchor, discriminative power, or a new metric. Table 1 positions this pilot against the closest locally verified work.

3 Experimental Protocol

3.1 Setup and configuration

We fix the training, initialization, generation, scoring, and signal-dump knobs in Table 2; Table 3 separately records the pre-registered multi-initialization control. This moves reconstructive values out of prose while preserving their scope and provenance.

3.2 Matched arms and training budget

Both arms use the converted SmolLM2-135M architecture (Ben Allal et al., 2025; Zhou et al., 2026). The AR-init arm retains pretrained weights; random-init is the fixed, audited from-scratch draw in Table 2. Crucially, the four paired repeats vary data order and training stochasticity, not this weight draw, so their loss variance cannot be transferred to independently drawn random initializations. To test that gap we ran a pre-registered multi-init negative control (Table 3) on three independently drawn random-initialization checkpoints, each trained under the same 30M-token / seed-42 data ordering as the two principal arms (Section 5). On the loss side the negative control reproduces: the paired last-logged-loss gap ($random - init_R - AR - init$) is positive on all three draws (bootstrap 95% CI lower bound > 0), so the sign-consistency of the principal result does not collapse across random initializations. The emission-side degeneration mode, however, is initialization-dependent: two draws degrade into a repeated-word lexical loop while the third is a shorter “gray” profile, so the emission narrative does not transfer across draws and is scoped to this effect (Section 5; evidence hash and verified fields are in `paper/anonymous_artifact/data/reproducibility_anchors.json`).

All token accounting, runtime, logging, objective, scheduler, hardware, and provenance fields are frozen in Table 2; in particular, “final” means the step-450 log inside the 458-step run, not a separate step-458 evaluation.

Table 2: **Controlled training and evaluation design.** Only the initialization checkpoint differs between the two principal arms. The PPL teachers and MAUVE configuration are evaluation choices, not independent quality anchors.

Training field	Frozen setting	Evaluation field	Frozen setting
Model / vocab	SmolLM2-135M converted to masked denoising; 49,152 base tokens plus $< mask >$ (49,153 total); tied embeddings	Checkpoint / prompt	seed-42 checkpoint pair; one prompt, "To be or not to be."
random-init draw	Linear/embedding weights i.i.d. $\mathcal{N}(0, (1/24)^2)$; LayerNorm gains 1, biases 0; host ambient RNG (no seed argument), hence one fixed draw; raw-HF A2D conversion uses <code>--random-init</code> (state dict skipped) before appending the mask row	Generation grid	$\tau \in \{0.7, 1.0\}$; 32 continuations per arm-temperature cell; 128 tokens; 128 denoising steps; block size 32; low-confidence remasking as recorded by the hashed sampler entry in the anonymous artifact
Init audit	Per-layer standard deviations within 0.1% of $1/24$; embedding normaltest $p = 0.92$, skew -0.004 , excess kurtosis 0.01 ($n = 20,000$); all 272 tensors differ; no step-0 scalar was logged, so the earliest step-10 losses, 9.6009 (A) and 11.2053 (B), are compared with $\ln(49,153) \simeq 10.802$	Length / PPL	Natural length plus caps $L \in \{16, 32, 48\}$ using $\min(n_{full}, L)$; prompt excluded; median PPL under GPT-2 (124M) and SmolLM2-135M
Data / budget	FineWeb-Edu sample/10BT shards 000/001/003/004; $16 \times 8 \times 458 \times 512 = 30,015,488$ tokens/arm; last logged value at step 450 of 458	MAUVE	<code>mauve-text==0.4.0</code> ; GPT-2-large features; holdout shard 005; $p = 100$, $q = 32$, <code>num_buckets=auto</code> $\rightarrow$ 3 buckets; <i>not recorded</i> in this pilot; two random-init cells identical at <code>frontier_integral=1.0</code>
Optimizer / hardware	AdamW; 10^{-4} peak LR; cosine; 10% warmup; bfloat16, ZeRO-2, $8 \times$ NVIDIA H20-class accelerators; reference pair 421 s / 0.94 GPU-h; three paired repeats 2.40 GPU-h	Emission / signal	Median emitted and alphabetic-word counts; corpus top-word share; schema v3.1 mask, entropy, max-probability, top- k , commit, final-argmax, and drift records
MDLM recipe	Uniform $t \in [10^{-3}, 1)$; linear α ; scheduler-weighted masked CE; <code>loss_norm_type=token</code> ; <code>right_shift_logits=False</code> ; implementation hashes in <code>paper/anonymous_artifact/data/reproducibility_anchors.json</code>	Argmax diagnostic	Most frequent final-step strict top-1 over $32 \times 128 = 4,096$ records; distinct from sampled-token frequency
Repeats / seed scope	Paired seeds 42, 2024, 2025, 2026 vary data order and training stochasticity on fixed initial weights; arm names were used during execution (label-coded, not human double-blind)	Provenance	Hashed driver, sampler, serialized training arguments, and exact regeneration pointers in <code>paper/anonymous_artifact/data/reproducibility_anchors.json</code> and Appendix A

Table 3: **Multi-random-init negative control** (pre-registered). Fresh seeds 11, 22, and 33 use the principal arms' 30M-token seed-42 data ordering. Loss gaps reproduce on all draws, whereas emission mode is initialization-dependent (two loops and one shorter "gray" profile).

Draw	Gap $B_R - A$ (nat)	Emission	Words ($\tau=0.7$)
<code>init-seed=11</code>	2.300	lexical loop	125.0
<code>init-seed=22</code>	2.281	lexical loop	122.5
<code>init-seed=33</code>	2.272	gray	35.0
bootstrap 95% CI lower bound	2.272 (> 0)	—	—

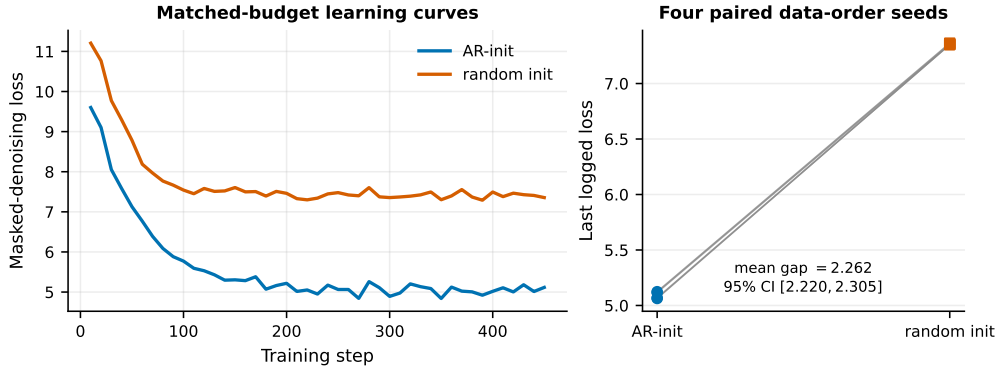


Figure 2: **Matched-budget training.** Left: both arms receive the same 458-step, 30.0M-token budget. Right: last-logged losses and paired-difference CI over four data-order seeds; connectors reveal overlapping points.

3.3 Generation and metric protocol

Table 2 gives the complete generation grid. It is a single-checkpoint, single-prompt, 32-sample-per-cell pilot rather than a multi-seed quality estimate. As preregistered, we report median causal PPL under GPT-2 and SmolLM2; the median prevents two catastrophic AR-init outliers at $\tau = 0.7$ from dominating. SmolLM2 is weight-related to AR-init and may share FineWeb-Edu source-domain bias with this experiment, so agreement with GPT-2 mitigates but does not remove teacher dependence.

Each PPL cell is accompanied by emitted length, alphabetic-word count, and a pooled corpus top-word share. The share is an illustrative cell aggregate; the artifact does not contain a per-continuation concentration distribution, so we make no continuation-level concentration claim. These fields describe the scored emission rather than its quality. The attempted MAUVE value and the final-step argmax diagnostic are retained under the same rule: both remain routine raw records, but neither is used for a pilot quality or capability inference. MAUVE is *not recorded* under Table 2, and strict top-1 remains separate from sampled-token frequency at nonzero temperature.

3.4 Frozen signal layer and probes

Schema v3.1 records mask state, entropy, maximum probability, top- k identities and probabilities, commit step, final argmax, and masked-position drift with explicit dtypes and shapes; evaluation labels remain outside the trajectory layer. One dump therefore supports unigram concentration, tie-aware unresolved-position drift, and committed-position sweep-tail analyses without another model run.

A fourth analysis applies one instrument to both 135M arms: among currently masked positions, entropy is scored against whether the current argmax differs from the eventual final token. Table 10 reports the raw AUROC, trajectory-bootstrap interval ($B = 2,000$, seed 0), trajectory and step counts, row support, and permutation calibration (Monte Carlo $B = 400$, seed 0). The calibration estimand is real/null95, where null95 is the 95th percentile of the permutation null; a value above one supplies at most a weak orientation check, not a quality label.

4 Results

4.1 AR initialization preserves a large short-horizon loss advantage

Figure 2 shows the learning curves, and Table 4 gives every paired endpoint. The gap $L_{\text{random}} - L_{\text{AR}}$ is positive for all four data-order seeds, with mean 2.262 nats and paired- t 95% CI [2.220, 2.305] ($n = 4$). The nominal one-sided sign test attains only its minimum $p = 0.0625$ at $n = 4$. These statistics vary data order and training randomness conditional on fixed initialization checkpoints, not independent weight draws.

Thus, under this recipe, autoregressive weights retain a roughly 2.26-nat short-horizon advantage after 30M tokens. Changing curves and one principal checkpoint per arm preclude convergence-time or asymptotic claims.

Table 4: **Paired last-logged loss endpoints.** All four differences have the same sign; mean $B - A = 2.2623$ nats, paired- t 95% CI [2.2196, 2.3050], and nominal one-sided sign-test $p = 0.0625$.

Data-order seed	AR-init	random-init	$B - A$
2024	5.0687	7.3501	2.2814
2025	5.0629	7.3522	2.2893
2026	5.1243	7.3637	2.2394
42	5.1159	7.3550	2.2391

Table 5: **Perplexity with required emission reporting.** PPL is the prompt-excluded median over 32 continuations. Length and content are median word counts; top-word share is pooled over the complete cell and is illustrative because no per-continuation distribution was retained. Short-stub counts report no-alphabetic-word and at-most-ten-word samples where audited. These coordinates describe the scored object, not a preferred arm.

Arm	τ	SmolLM2 PPL	GPT-2 PPL	Emitted length	Content words	Top word share	Short-stub counts
							9 no-alpha
AR-init	0.7	10.09	9.08	5.0	1.0	and / 38.0%	$25 \leq 10$
random-init	0.7	3.50	5.47	126.0	126.0	of / 56.8%	—
AR-init	1.0	18.96	25.91	8.5	3.5	and / 17.5%	—
random-init	1.0	15.08	20.44	117.0	115.5	of / 46.9%	—

Table 6: **Within-arm before/after discrimination check.** Same prompt, seed, τ , and protocol; only training state differs. PPL is the SmolLM2 median and concentration fields use the complete emitted cell.

Coordinate	Untrained (0-step)	Trained (458-step)
SmolLM2 median PPL	237,873.60	3.50
Median emitted words	82	126
Alphabetic-word yield	0.991	1.000
Top word / share	crossings / 0.42%	of / 56.79%
Top-five cumulative share	1.25%	98.16%
Distinct-token ratio	0.941	0.023

4.2 The degenerate arms differ on emission coordinates

Table 5 reports the object scored by every perplexity cell. With no independent quality anchor, the natural-output medians do not identify a preferred arm. The accompanying fields establish only that the two degenerate arms differ on coordinates outside this pilot’s PPL entry; its attempted MAUVE entry is not recorded. At $\tau = 0.7$, AR-init is predominantly a short punctuation/newline remnant, whereas random-init continues in a single-word lexical loop; the complete counts and shares appear in Table 5.

A within-arm discrimination check. Table 6 compares random-init before and after 458 steps under the same prompt, seed, $\tau = 0.7$, and 32×128 protocol. Teacher PPL moves by nearly five orders of magnitude, while the untrained draw is a scattered high-entropy stream and the trained checkpoint a single-token loop. The PPL entry alone does not encode the loop, while the emission fields do not encode teacher fluency. This one-arm, one-prompt, one-checkpoint-pair check establishes complementary, non-redundant information only on that pair, not cross-initialization, cross-prompt, or cross-checkpoint discrimination power.

For a common-length view, Table 7 caps each sample at $\min(n_{\text{full}}, L)$ while retaining shorter samples at natural length. The gap survives each SmolLM2 cap, nearly collapses for GPT-2 at $L = 32$, and can reverse at $\tau = 1.0$. Only one of 32 AR-init, $\tau = 0.7$ samples reaches 128 SmolLM2 tokens; its teacher-dependent score is not an estimate. Hence the ordering is not robust to teacher, cap, or temperature and supports no robust quality ranking.

The frozen MAUVE configuration is listed in Table 2. Its two random-init cells are numerically identical at the raw frontier integral, so MAUVE is *not recorded*: no ratio, quality ordering, or capability claim is formed, while the raw values remain in the artifact for reproduction.

The concentration measurements describe the emitted object rather than an inferred quality label (Table 8). In random-init, the strict final-step top-1 is nearly a single token, while per-position predictive

Table 7: **Length-capped median PPL at $\tau \in \{0.7, 1.0\}$.** AR-init and random-init use $\min(n_{\text{full}}, L)$ teacher tokens; “full” is the natural $\tau = 0.7$ median. Gap is AR-init minus random-init.

SmolLM2-135M teacher					GPT-2 teacher				
τ	Cap	AR-init	random-init	Gap	τ	Cap	AR-init	random-init	Gap
0.7	full	10.09	3.50	+6.59	0.7	full	9.08	5.47	+3.61
0.7	16	16.31	9.54	+6.77	0.7	16	39.54	14.52	+25.0
0.7	32	9.93	5.29	+4.64	0.7	32	9.39	8.69	+0.70
0.7	48	10.09	4.31	+5.78	0.7	48	9.08	5.92	+3.16
1.0	16	58.19	54.67	+3.52	1.0	16	104.83	71.00	+33.8
1.0	32	19.72	35.56	-15.8	1.0	32	44.50	47.09	-2.59
1.0	48	23.54	23.21	+0.33	1.0	48	37.47	33.28	+4.19

Table 8: **Argmax concentration and commitment coordinates.** Final-step strict top-1 covers 4,096 positions ($\tau = 0.7$, seed 42, 128 steps); agreement is zero-tolerance integer-id equality between the committed token and strict final-step top-1.

Coordinate	AR-init	random-init
Records with top-1 = token 260 (the)	—	4,094 / 4,096
Final-step top-1 entropy (bits)	3.00	0.006
Final-step top-1 ties	present	zero
Top word in $\tau = 0.7$ emission	and / 38.0%	of / 56.8%
Per-position predictive entropy (nats)	sweep tail (Fig. 3)	≈ 7.2 –8.1 (flat)
Committed-vs-top-1 agreement	0.467	0.394

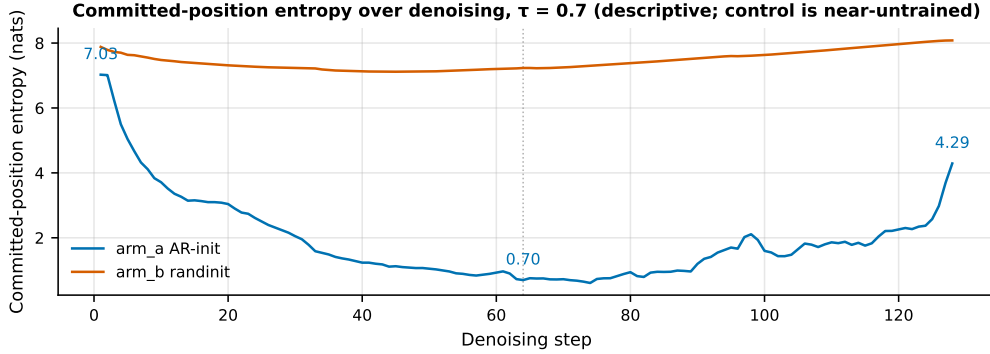


Figure 3: **Committed-position entropy over denoising.** The AR-init arm develops a mid-trajectory minimum and high-entropy residue; random-init remains high. Curves use v3.1 stored-entropy fields at $\tau = 0.7$ and do not use the failed scalar/top- k drift recomputation.

entropy remains high throughout — so a “de-facto argmax filter” reading of the committed stream is not supported by the data: the committed token can disagree with the strict final-step top-1, and argmax concentration and the emitted distribution decouple.

4.3 Trajectory records under a failed drift recomputation gate

The probes give three complementary views. First, random-init has the more concentrated final-step argmax (Table 8). Second, the original claim that low-confidence layers cause drift is unsupported: the failed scalar/top- k gate leaves no quantitative error bound for drift localization. The passing tie-bound and decoding-alignment checks establish only that the serialization is internally bounded and aligned; they do not recover the failed counts or a directional mechanism. Third, AR-init falls from high entropy to a mid-trajectory minimum and rises to a high-entropy residue, whereas random-init stays nearly flat (Figure 3). This stored-entropy contrast does not use the failed drift field, but the nearly untrained control (458 steps, final loss 7.36) still cannot distinguish learned structure from the masking schedule.

Table 9 reports the failed field and the two passing serialization checks; neither supplies a drift effect or error bound.

Table 9: **Drift recomputation gate.** Specified top- k recomputation versus scalar totals at 1% tolerance. The failed fields are shown only to diagnose the gate; the passing fields support serialization alignment, not a drift effect estimate.

Quantity	AR-init	random-init
Aggregate scalar/top- k difference	12.6%	18.1%
Worst single-record relative error	56.8% ($57\times$ tolerance)	—
Tie-bounded consistency gate	pass	pass
Decoding-alignment gate	pass	pass

Table 10: **Masked-entropy orientation, support, and calibration.** Raw AUROC labels whether current argmax differs from the final token; CIs use trajectory bootstrap ($B = 2,000$, seed 0). Calibration uses 400 seed-0 permutations and is real/null95, where null95 is the permutation-null 95th percentile; it is not an AUROC. NR denotes a field absent from the frozen local artifact.

Bed	Traj.	Steps	Masked n	AUROC [95% CI]	Calib. multiple
AR-init (135M)	32	128	260,064	0.686 [0.652, 0.725]	1.334
random-init trained (135M)	32	128	260,064	0.458 [0.453, 0.463]	1.074
random-init untrained bed	NR	NR	NR	—	1.009

4.4 Masked-entropy orientation check

The two 135M masked-position entropy results are collected in Table 10. The previously explored 8B row is omitted because its checkpoint, prompts, and sampling temperature are not fully specified in the anonymous artifact. Off-axis calibration uses a near-one untrained-bed multiple as a null check. The raw random-init AUROC is 0.45785, with a roughly 0.010-wide interval over a pool of 260,064 dependent masked-position rows; the resampling unit remains only 32 trajectories. We therefore read it as a weak, near-null negative orientation, not as strong anti-predictive behavior or an inverse quality signal. The AR-init contrast is 0.68593 [0.6520, 0.7255], also descriptive rather than a quality score.

5 Discussion

A negative control is a reporting unit test. The random-initialized arm is a unit test for describing distinct degeneration modes, not a competitive baseline. For this single 458-step, single-checkpoint pilot, the within-arm check in Section 4.2 supports only this companion bundle:

$$(\text{median teacher PPL, distributional similarity, } \mathcal{E}), \quad (1)$$

where $\mathcal{E}$ is effective length, alphabetic yield, and emitted-token concentration. The attempted MAUVE value and argmax diagnostic follow the same retention standard: raw records remain, but neither supports inference. Slot (ii) is therefore unexercised; argmax is not an emission slot because it decouples from sampled emissions here (Section 4.2, Figure 1). These fields are not a new metric, and the multi-init control supports only the loss-side sign, not the PPL/loss ordering.

Loss calibration and generation quality answer different questions. The paired loss gap is statistically repeated; generation uses only seed-42 checkpoints and diagnoses emission coordinates, not production quality. Neither implies good text, a MAUVE gap, or a quality ranking for AR-init.

The signal layer is infrastructure, not a new decoder. The schema separates masked from committed positions, current argmax from held token, and top- k ties. It diagnoses this control; adaptive-decoder accuracy or efficiency remains outside the evidence.

6 Limitations

Table 11 pairs each frozen boundary with its inferential ceiling.

Ethics and AI assistance. No human subjects are involved. The web corpus retains privacy, licensing, and representational-bias risks. AI assisted reorganization and copyediting; quantitative claims remain artifact-bound and the authors responsible.

Table 11: **Frozen scope and interpretation boundaries.** Complete settings are in Table 2; each row states its inferential ceiling.

Axis	Frozen support	Interpretation boundary
Training horizon	30M tokens/arm; losses still evolving; last log at step 450 of 458	No convergence-time speedup or asymptotic ordering.
Paired variation	Four data-order/training seeds with fixed initial weights	The paired CI does not estimate random-weight-draw variance.
Multi-init control	Three fresh draws (seeds 11, 22, 33): positive loss gap on all three; two lexical loops and one shorter “gray” emission	Loss-side sign consistency survives this low-power three-draw pilot; emission mode is initialization-dependent. This forward analysis is non-gating and not a headline.
Generation	135M bed; seed-42 checkpoint pair; one prompt; $\tau \in \{0.7, 1.0\}$; 32 samples/cell; 128-step sampler	No extrapolation to other prompts, checkpoints, dLLMs, or LLaDA-8B.
PPL / MAUVE	Length-confounded teacher PPL; one full-length AR-init sample; source-related SmolLM2 teacher; pinned <code>mauve-text==0.4.0</code> but automatic three-bucket value not recorded	Ordering is teacher-, temperature-, and length-construction-dependent; no MAUVE capability or quality conclusion.
Drift / causality	Scalar/top- k totals differ by 12.6–18.1%; all probes observe one sampler	No quantitative error bound, drift effect estimate, decomposition, or causal mechanism without a corrected rerun.
Upgrade paths	Only three independent random-weight draws; drift gate remains failed	More weight draws and a corrected drift recomputation are future evidence upgrades, not results or conditions claimed here.

7 Conclusion

In this 135M negative-control pilot, autoregressive initialization retains a 2.262-nat last-logged loss advantage across four paired data orders, while one seed-42 generation check contrasts two degenerate emissions. Effective length, content yield, and concentration distinguish them; PPL is recorded but the pinned MAUVE attempt is not. Without an independent quality anchor, this is only a companion record for this setting—not a quality ordering, metric capability, new metric, training discovery, or extrapolation to other dLLMs.

Reproducibility Statement

The anonymous bundle `paper/anonymous_artifact/` has manifest SHA-256 `10bde5412c17f2e65b0ea1230f8a2b2db134c02b6eda8b92e560211bbae6e735`. It contains emission data, four-seed endpoints, implementation hashes, regeneration pointers, and the pre-registered multi-init summary (source SHA-256 `546af24c0267ceba08529081c0425d9f370e0ea0d02869dc3c98ad490905c1ee`). No verified external anonymous URL is available at this freeze. The seed-42 principal safetensors checkpoints have full SHA-256 values AR-init `a618886a762eb8983cc0821fb43261ff27f4012738e240d00827d0a81d7dd00d` and random-init `bbfbfb7fa4782dea8492412edd9bf30eedc1940da599848bf86d39b8c942a509`. Table 2 records the random draw’s distribution and scale; its ambient weight seed is missing and differs from training/data-order seed 42. Step 0 was not logged; the earliest step-10 losses are 9.6009 and 11.2053, respectively, versus $\ln(49,153) \simeq 10.802$.

Source commit `6f21ee9` builds with Tectonic 0.17.0, the pinned ICLR 2026 template, and fixed `SOURCE_DATE_EPOCH`; the rebuild is recorded in `paper/layout/deterministic.json`. Runtime pins are `transformers==4.57.0`, `datasets==4.2.0`, `accelerate==1.11.0`, `sentencepiece==0.2.0`, and `mauve-text==0.4.0` on PyTorch/bfloat16, ZeRO-2, and eight anonymized H20-class accelerators. The package pin does not make `num_buckets=auto` → 3 byte-reproducible, so the attempted MAUVE value remains *not recorded*. Trajectory CIs use $B = 2,000$, seed 0; permutation calibration uses $B = 400$, seed 0, with null95 the permutation-null 95th percentile. All anchors are in `paper/anonymous_artifact/data/reproducibility_anchors.json`.

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A Reproducibility and Evidence Index

Table 12 maps every headline claim to an on-disk artifact; the companion `paper/anonymous_artifact/README.md` gives field-level JSON pointers, commands, and scope restrictions.

Table 12: **Claim-to-artifact evidence map.** Each headline claim is bound to a primary on-disk artifact and its regeneration or audit entry; `paper/anonymous_artifact/README.md` carries the field-level JSON pointers.

Claim	Primary artifact	Regeneration / audit
Training budget, curves, paired gap, and CI	<code>paper/anonymous_artifact/data/multiseed_summary.json</code>	<code>scripts/multiseed_postprocess.py</code> ; <code>data/reproducibility_anchors.json</code>
PPL, degraded MAUVE, emission coordinates, and length checks	<code>paper/anonymous_artifact/data/phase2_summary.json</code> ; <code>emission_coords.json</code>	<code>scripts/phase2_emission_coords.py</code> ; <code>scripts/phase2_gen_ppl_lengthmatched.py</code>
Argmax, drift, and sweep-tail probes	<code>paper/anonymous_artifact/data/unigram.json</code> ; <code>drift_layer.json</code> ; <code>sweep_tail.json</code>	Frozen v3.1 fields; drift failure retained as a gate diagnostic
Masked-entropy AUROC	<code>paper/anonymous_artifact/results/cross_paradigm.json</code>	<code>data/reproducibility_anchors.json</code>